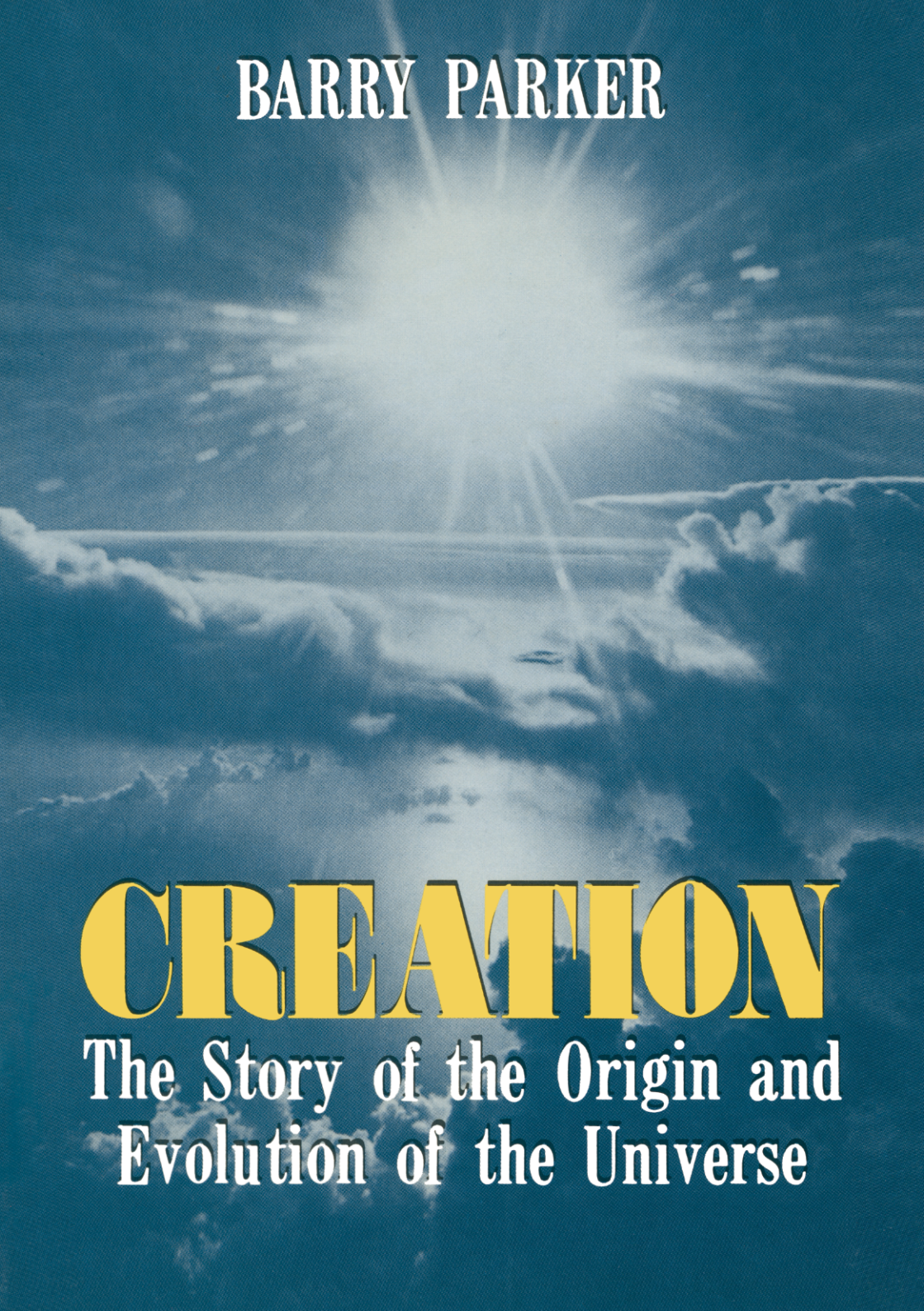

BARRY PARKER
CREATION
The Story of the Origin and Evolution of the Universe

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CHAPTER 17

Epilogue

Our story is now complete: we have traced the universe from its beginning to the formation of life. The picture we have presented, most astronomers would agree, is fairly accurate. Yet many questions remain unanswered. Some of our ideas, for example the explanation of the lack of antimatter in the universe, are based on the existence of a grand unified theory. There are several versions of this theory, and we are not yet certain which, if any, is correct. Indeed, it is now quite possible that all grand unified theories will be superseded by a superstring theory, and superstring theories may give us a quite different picture of creation. Without a doubt, though, if we are to thoroughly understand creation we must first completely understand the particles and forces of the universe. And we can get this understanding only if new, larger particle accelerators such as the superconducting supercollider (SSC) are built. It is also important that larger telescopes be built, and some of them be put into space.

But what will happen if new tests show us that some of our ideas are wrong? Obviously we will have to start over and look for a better theory. It is, in fact, possible that this will happen. Inflation theory may eventually be shown to be incorrect. It is even possible that the big bang theory is wrong. Only time will tell.

Turning now to the other end of the story, there is still the question of what will eventually happen to the universe. For an

answer we must look again at Friedmann's theory. It tells us that the universe will either expand forever, or stop expanding and collapse back on itself, depending on its average density. If the average density is over a certain critical amount, the mutual gravitational pull of the matter will eventually stop the expansion. If not, the universe will expand indefinitely. We are still not certain which of these will happen, but as we saw earlier, the density appears to be very close to the dividing line.

Before I end the book I feel I should address a question that some of you will no doubt be wondering about: Why does a book on the creation of the universe have no mention of God? Scientists do, indeed, rarely mention God when they talk about creation. Furthermore, they are sometimes accused of trying to do away with the need for a God by attempting to explain creation in scientific terms. And it is true: scientists would prefer a purely scientific explanation of the beginning of the universe. That is not to say, though, that all scientists are atheists (few are). Furthermore, there is no fear that scientists will ever eliminate the need for a God. If we look back at the early universe we see that regardless of how far things are pushed—even if we were someday able to explain creation itself in an entirely satisfactory scientific way—there is still something that is unexplained. Creation depends on the basic laws of nature—without them it would not be possible. Who created these laws? There is no question but that a God will always be needed.